



Computer Networks

IT351

Project-Phase 2

Deadline: Day 09/08/2026 @ 23:59

[Total Mark is 15]

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instruction

Project Objective:

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

Group Details

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

Project Report

- The project is divided into two phases:
 - Phase 1 (Mid project Progress Report)
 - Phase 2 (Final project Report)
- This report is for Phase 2. In this report, you should complete all the requirements for Phase 2.
- Each Report will be evaluated according to the marking criteria

*Learning
Outcome(s):

Demonstrate
protocol
configuration,
network-addressin
g schemes and
analyze packet
transmission for
the Local and
Wide Area
Networks.*

Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

Network Design Requirements:

1. WAN 1 (Main Campus):

- Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
- Connect each LAN to a router (Router 1).

2. WAN 2 (Branch Campus):

- Contains LAN 3 (Library) and LAN 4 (Faculty Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
- Connect each LAN to a router (Router 2).

3. WAN Connection:

- Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
- Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

Configuration Details:

1. Assign IP Addresses:

- Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
 - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).

- LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).
- LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).
- LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. Router Configuration:

- Configure IP addresses on the LAN and WAN interfaces of both routers.

Example:

- Router 1:
 - LAN 1 Interface: 172.16.10.1/24
 - LAN 2 Interface: 172.16.20.1/24
 - WAN Interface (Serial): 10.10.10.1/30
- Router 2:
 - LAN 3 Interface: 172.16.30.1/24
 - LAN 4 Interface: 172.16.40.1/24
 - WAN Interface (Serial): 10.10.10.2/30

3. Enable Routing:

- Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.
-

Phase 2 Requirements

Testing and Verification

1. Use the ping command to test connectivity between a device in any LAN of WAN 1 and a device in any LAN of WAN 2.
2. Verify that data packets are successfully transmitted (0% packet loss).

Questions to Answer (10 Marks — 2 marks each)

Answer all five questions. For each question, you must include (a) a clear screenshot from your Packet Tracer project and (b) the commands used at that instance (e.g., IOS CLI commands or the configuration steps applied). Screenshots without the corresponding commands/steps will receive ZERO marks for that question.

1. **Creating nodes:** Show a screenshot of the complete topology with all nodes created (12 end devices, 4 switches, 2 routers). List the device types selected and any steps/commands used to name the devices (e.g., hostname configuration). (2 marks)
2. **Giving connections in the LAN:** Show a screenshot of the LAN connections between end devices, switches, and routers. State the cable types used for each connection and why. (2 marks)
3. **Assigning IP addresses:** Show screenshots of IP address, subnet mask, and default gateway assignment on at least one end device in each LAN, and the commands used to configure the router LAN interfaces (e.g., interface, ip address, no shutdown). (2 marks)
4. **Creating the WAN connection:** Show a screenshot of the serial link between Router 1 and Router 2, and the commands used to configure the serial interfaces and enable routing (static routes or a dynamic routing protocol such as RIP), including clock rate if applicable. (2 marks)
5. **Successful data transmission using the ping command:** Show screenshots of successful ping tests between a device in WAN 1 and a device in WAN 2 (both directions), with the exact ping commands used and the results (replies received, 0% loss). (2 marks)

Written Reflection — End of Project Report (5 Marks)

The project report must include a Reflection section (300–500 words) as the last section before the Project Artifacts section. This section must be written in the students' own words and should address some of the following:

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method (e.g., static routing vs. RIP, your IP addressing scheme) over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

- Live report document (with edit history): [link]
- GitHub/GitLab repository (with commit history): [link]
- Packet Tracer project file (.pkt): [link or attached]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The final report must be submitted as a PDF, together with the live document link, the repository link, and the Packet Tracer (.pkt) file.
- The edit history of the live document and the commit history of the repository must show incremental progress across both phases (Phase 1 and Phase 2).

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Network Topology

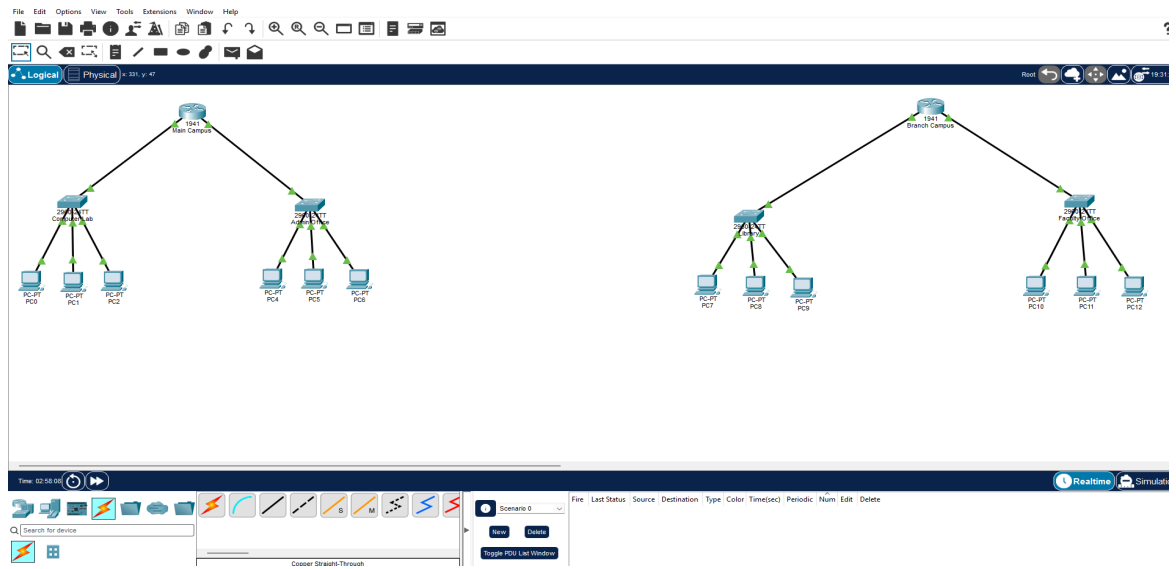


Figure 1. Initial Network Topology

Devices Used in the Network

Device Type	Model	Quantity
Router	Cisco 1941	2
Switch	Cisco 2960-24TT	4
PC	PC-PT	12

Commands Used to Change Device Names

Router (**Main Campus**):

```
enable
configure terminal
hostname MainCampus
end
copy running-config startup-config
```

Router (**Branch Campus**):

```
enable
configure terminal
hostname Branch Campus
end
copy running-config startup-config
```

Switch (**Computer Lab**) :

```
enable
configure terminal
hostname ComputerLab
end
copy running-config startup-config
```

Switch (**Admin Office**):

```
enable
configure terminal
hostname AdminOffice
end
copy running-config startup-config
```

Switch (**Library**):

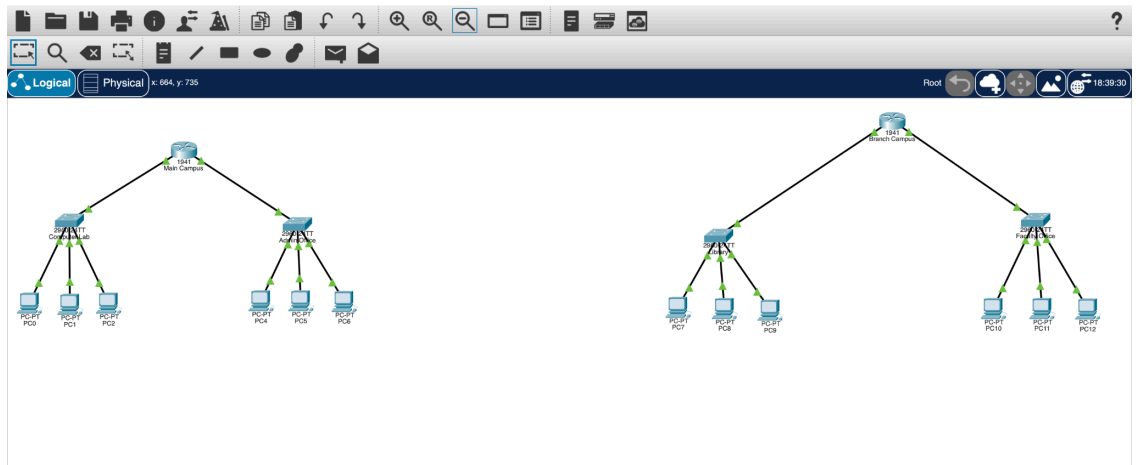
```
enable
configure terminal
hostname Library
end
copy running-config startup-config
```

Switch (**Faculty Office**):

```
enable
configure terminal
hostname FacultyOffice
end
copy running-config startup-config
```

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Connection	Cable Type	Reason
PC to Switch	Copper Straight-Through	Used to connect different types of devices (PC and Switch).
Switch to Router	Copper Straight-Through	Used to connect different types of devices (Switch and Router).

LAN	Network	Subnet Mask	Default Gateway
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ComputerLab 172.16.10.0/24 255.255.255.0 172.16.10.1

AdminOffice 172.16.20.0/24 255.255.255.0 172.16.20.1

Library 172.16.30.0/24 255.255.255.0 172.16.30.1

FacultyOffice 172.16.40.0/24 255.255.255.0 172.16.40.1

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The image displays two screenshots of a network configuration interface, likely from a Packet Tracer simulation. Both screenshots show the 'IP Configuration' tab for a specific PC.

PC4 Configuration:

- Interface: FastEthernet0
- IP Configuration:
 - DHCP: ☐ Static: ☒
 - IPv4 Address: 172.16.20.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 172.16.20.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:
 - Automatic: ☐ Static: ☒
 - IPv6 Address: [Empty field] / [Empty field]
 - Link Local Address: FE80::207:ECFF:FE4E:6622
 - Default Gateway: [Empty field]
 - DNS Server: [Empty field]
- 802.1X:
 - Use 802.1X Security: ☐
 - Authentication: MDS
 - Username: [Empty field]
 - Password: [Empty field]

PC0 Configuration:

- Interface: FastEthernet0
- IP Configuration:
 - DHCP: ☐ Static: ☒
 - IPv4 Address: 172.16.10.3
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 172.16.10.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:
 - Automatic: ☐ Static: ☒
 - IPv6 Address: [Empty field] / [Empty field]
 - Link Local Address: FE80::201:43FF:FE14:4030
 - Default Gateway: [Empty field]
 - DNS Server: [Empty field]
- 802.1X:
 - Use 802.1X Security: ☐
 - Authentication: MDS
 - Username: [Empty field]
 - Password: [Empty field]

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PC7

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.16.30.2

Subnet Mask 255.255.255.0

Default Gateway 172.16.30.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::20C:FFFF:FE8B:A64

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC10

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.16.40.2

Subnet Mask 255.255.255.0

Default Gateway 172.16.40.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2E0:80FF:FE9C:298A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

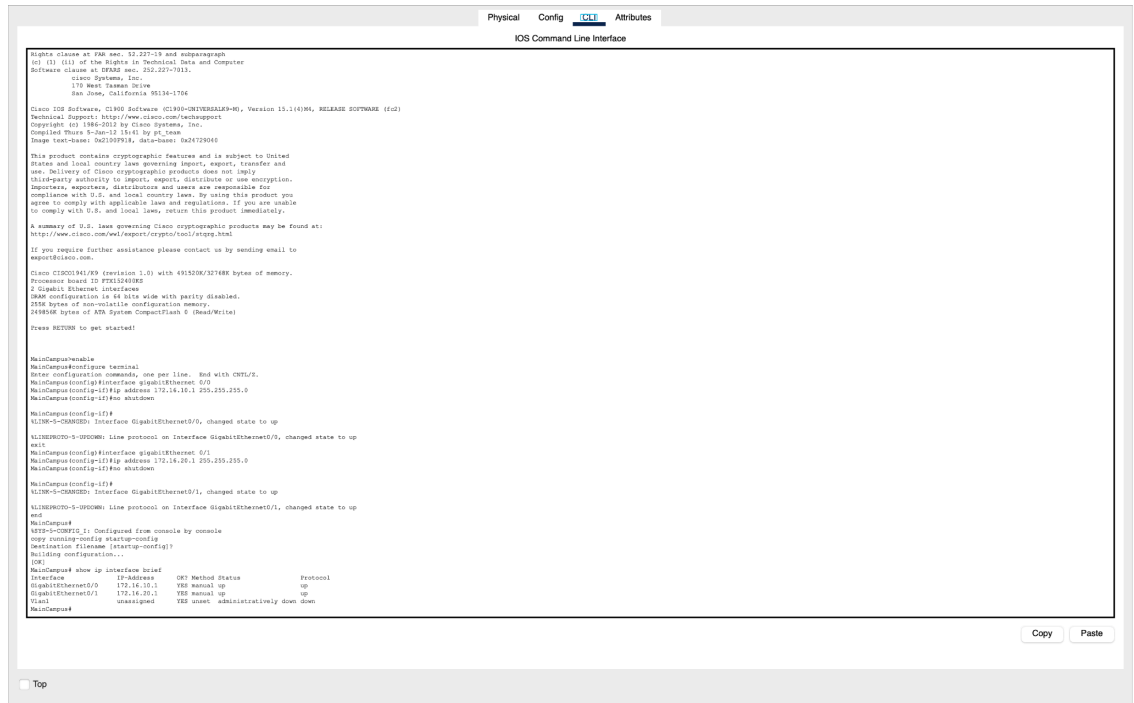
Authentication MD5

Username

Password

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enable

configure terminal

interface gigabitEthernet 0/0

ip address 172.16.10.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

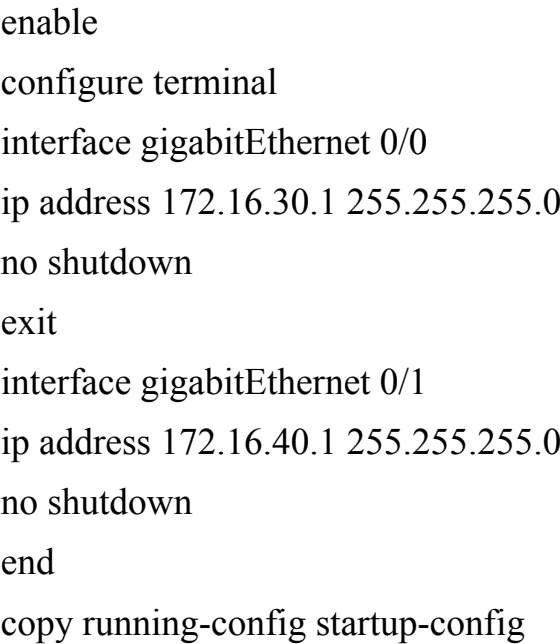
ip address 172.16.20.1 255.255.255.0

no shutdown

end

copy running-config startup-config

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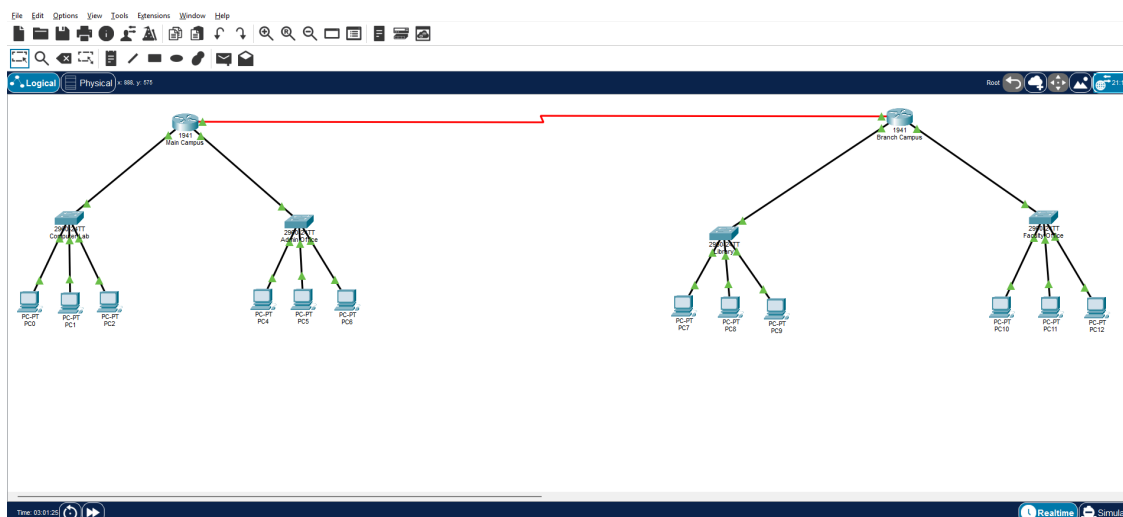


Figure - Final Design of the Network

Commands for configuring the serial interface and enabling routing:

```

Main Campus
Physical Config CLI Attributes
Cisco IOS Command Line Interface

MainCampus> enable
MainCampus> configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
MainCampus(config)# interface Serial0/1/0
MainCampus(config-if)# ip address 10.10.1.255 255.255.255.252
MainCampus(config-if)# clock rate 56000
MainCampus(config-if)# no shutdown
MainCampus(config-if)# exit
MainCampus(config)#
%LINK-3-UPDOWN: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

MainCampus con0 is now available

MainCampus>

MainCampus> enable
MainCampus> configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
MainCampus(config)# ip route 172.16.0.0 255.255.255.0 10.10.10.2
MainCampus(config)# ip route 172.16.40.0 255.255.255.0 10.10.10.2
MainCampus(config)#
MainCampus>
RTB-S-CORP01_1_ Configured from console by console
  
```

Router 1 (**Main Campus**):

/Serial interface configuration

enable

configure terminal

interface serial0/1/0

ip address 10.10.10.1 255.255.255.252

clock rate 64000

no shutdown

exit

/Static routes

ip route 172.16.30.0 255.255.255.0 10.10.10.2

ip route 172.16.40.0 255.255.255.0 10.10.10.2

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```

Branch Campus
Physical Config CLI Attributes
IOS Command Line Interface

1 GigabitEthernet interfaces
2 200-based serial (sync/async) network interfaces
3 ROM configuration is 64 bits wide with parity disabled.
4 256K bytes of non-volatile configuration memory.
5 2498160 bytes of ATA System CompactFlash 0 (read/write)

Press RETURN to get started!

VLINEPOTD-4-UDONM: Line protocol on Interface GigabitEthernet0/0, changed state to up
VLINEPOTD-4-UDONM: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router#config hostname BranchCampus
BranchCampus(config)# interface serial10/1/0
BranchCampus(config-if)# ip address 10.10.10.2 255.255.255.252
BranchCampus(config-if)# no shutdown
BranchCampus(config-if)#
VLINEPOTD-4-UDONM: Interface Serial10/1/0, changed state to up
VLINEPOTD-4-UDONM: Line protocol on Interface Serial10/1/0, changed state to up

BranchCampus con0 is now available

Press RETURN to get started.

BranchCampus# enable
BranchCampus# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
BranchCampus# configure terminal
BranchCampus# ip route 172.16.10.0 255.255.255.0 10.10.10.1
BranchCampus# ip route 172.16.10.0 255.255.255.0 10.10.10.1
BranchCampus#
BranchCampus#
K979-9-C9790_1: Configured from console by nmcnle

```

Router 2 (**Branch Campus**):

/Serial interface configuration

enable

configure terminal

interface serial10/1/0

ip address 10.10.10.2 255.255.255.252

no shutdown

exit

/Static routes

ip route 172.16.10.0 255.255.255.0 10.10.10.1

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```
ip route 172.16.20.0 255.255.255.0 10.10.10.1
```

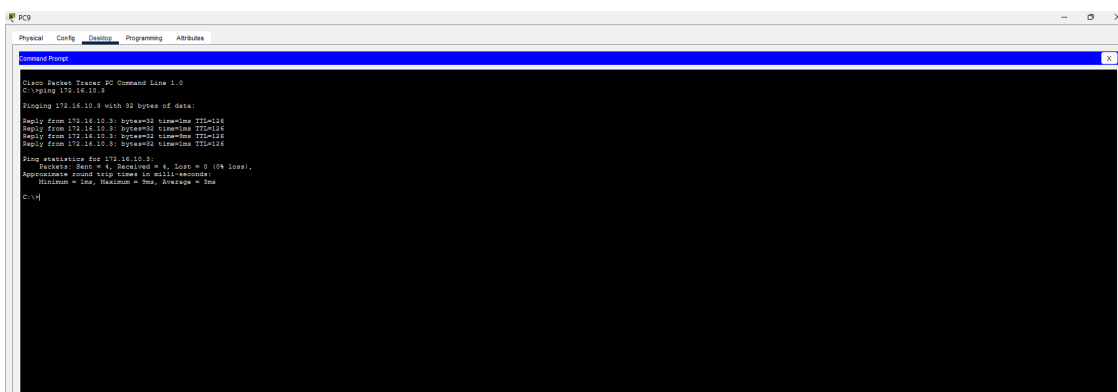
Data Transmission using the Ping Command:



```
PC4
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 172.16.30.2
Pinging 172.16.30.2 with 32 bytes of data:
Reply from 172.16.30.2: bytes=32 time=1ms TTL=128
Reply from 172.16.30.2: bytes=32 time=1ms TTL=128
Reply from 172.16.30.2: bytes=32 time=1ms TTL=128
Reply from 172.16.30.2: bytes=32 time=1ms TTL=128
Ping statistics for 172.16.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>
```

Figure - From Main Campus to Branch Campus

```
ping 172.16.30.2
```



```
PC9
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 172.16.10.3
Pinging 172.16.10.3 with 32 bytes of data:
Reply from 172.16.10.3: bytes=32 time=1ms TTL=128
Reply from 172.16.10.3: bytes=32 time=1ms TTL=128
Reply from 172.16.10.3: bytes=32 time=1ms TTL=128
Reply from 172.16.10.3: bytes=32 time=1ms TTL=128
Ping statistics for 172.16.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>
```

Figure - From Branch Campus to Main Campus

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ping Version History / Commit Log

The following internal record summarizes the main stages represented in the submitted report and Packet Tracer file.

Version	Phase	Work Completed	Status
1.0	Phase 1	Created the initial topology, selected devices, and named routers and switches.	Completed
1.1	Phase 1	Connected LAN devices and prepared the IPv4 addressing plan for all four LANs.	Completed
2.0	Phase 2	Configured LAN interfaces, serial addressing, clocking, and static routes.	Completed
2.1	Phase 2	Verified bidirectional communication with successful ping tests and 0% packet loss.	Completed
2.2	Final	Reviewed commands, added reflection and artifacts, and prepared Word and PDF files.	Completed

172.16.10.3

Reflection

Completing this project changed the way we approached network troubleshooting because every part of the design had to work as one system. The most significant challenge was establishing reliable communication between the Main Campus and the Branch Campus after the four LANs had been configured. Local communication was straightforward, but the first cross-WAN tests required careful checking of router interfaces, the serial link, default gateways, and routing entries. We reviewed the addressing table, used the router interface status and configuration outputs, and checked the path step by step. The serial connection was enabled on both routers, a clock rate of 64000 was applied on the DCE side, and each required static route was verified. We then repeated the ping tests in both directions. Receiving four replies with 0% packet loss confirmed that the LAN and WAN configurations were working together correctly.

We selected static routing rather than RIP because the topology is small and contains only two routers and four known LANs. Static routes made the forwarding path clear and allowed every destination network and next-hop address to be checked directly. RIP would reduce manual updates if the network grew, but it would introduce unnecessary routing advertisements and additional configuration for this limited design. We also used /24 networks for the LANs because they were easy to organize and distinguish, while the /30 subnet on the serial link provided only the two usable addresses required for the point-to-point connection. The main tradeoff is that static routing is less scalable and would require manual changes if more campuses or networks were added.

AI assistance was used only to proofread the final wording, review document formatting, and identify small presentation issues in the listed commands. It was not used to operate Packet Tracer or replace the group's configuration work. Every technical correction was checked against the actual topology, router outputs, IP settings, and successful ping results before it was included.

If we repeated the project, we would test and document each stage immediately after completing it. We would save clearer screenshots of interface status, routing tables, and ping results, use a structured troubleshooting checklist, and keep more frequent local versions of the report and Packet Tracer file. This would make errors easier to isolate and produce a stronger final submission.

Project Artifacts

Cisco Packet Tracer project file (.pkt): IT351_FinalDesign.pkt (attached separately).

Final written report: IT351_Project-Phase_2_Final.docx.

Final PDF report: IT351_Project-Phase_2_Final.pdf.

Google Doc: [Project-Phase2](#)

GitHub/GitLab repository: [GitHub](#)

Packet Tracer project file (.pkt): [Packet Tracer](#)

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Final Submission

Submission Item	Status
Final Word report (.docx)	Prepared
Final PDF report (.pdf)	Prepared
Cisco Packet Tracer project (.pkt)	Attached separately